Module 5 Assignment — Computational Lab 5

Start Assignment

- Due Thursday by 6pm
- Points 100
- Submitting a text entry box or a file upload
- Available Oct 8 at 12am - Dec 20 at 11:59pm

Module 5 Assignment: Variant Interpretation on Your Own Gene

Computational Lab 5 · Fall 2026

Weight: 100 points · Computational Lab 5 (3% of course grade)

Due Thursday, October 22 at 6:00 pm (start of Week 7). Complete lab Parts 1 and 2 first.

What this assignment is

In the Module 5 lab you built four variant predictors from scratch on a toy gene, then ran the real ones across five pharmacogenes together. Now you apply those skills to one gene of your own: you fetch its variants yourself, work out which tools can see them, calibrate a score into evidence, and classify the uncertain ones. You finish with a defensible provisional call for each variant of uncertain significance, backed by evidence you can quote.

Maps to course outcomes: LO1 (apply CADD, REVEL, AlphaMissense and the AlphaGenome Atlas), LO3 (interpret results in biological and clinical context), LO4 (evaluate calibration, independence and limits), LO5 (professional documentation).

Before you start

- Complete lab Parts 1 and 2. This assignment reuses their code: fetching from MyVariant.info, parsing the nested JSON, the coverage audit, and calibrate(). Part 2's commentary explains every step you are asked to repeat.
- Environment: biot6900 with numpy, pandas, matplotlib, scikit-learn and requests, plus an internet connection for the first run. Everything caches to data/ afterwards.
- Optional: an AlphaGenome API key for Step 7. Free for non-commercial use, about five minutes to set up, and Section 7 of the Module 5 Lab Guide has the exact commands. If you cannot get one, set SKIP_ATLAS = True and no marks are lost.
- Files: the assignment notebook BIOT6900_Module5_Part3_Assignment.ipynb (Canvas), and your assigned gene from the allocation list on Canvas.

What to do

Complete each TODO cell in the assignment notebook, following the steps below.

Step 1 · Set your gene

Set MY_GENE in the settings cell to the gene assigned to you on Canvas: one of CYP2C19, CYP2C9, VKORC1, DPYD or TPMT. If you were not assigned one, pick any and say which in your README.

The five genes are not equally easy, and that is intentional. VKORC1 in particular is small and may not have enough labelled variants to calibrate. If that is your gene, recognising and reporting the problem is the task, not working around it.

Step 2 · Fetch and quality-check your gene (15 pts)

Use fetch_gene() to pull every ClinVar variant in your gene from MyVariant.info, then cache it to data/<YOUR_GENE>.tsv so later runs do not re-download. Pass the result through numeric() and add a label column with clinvar_label().

Then run three quality checks and report each one. An incomplete fetch does not raise an error; it silently ruins everything downstream.

- How many variants came back? A handful means your query string is wrong.
- Did the known star alleles for your gene appear? ANCHORS lists them. If yours is missing, stop and fix the fetch before continuing.
- **Score columns.** Are CADD, REVEL and gnomAD populated at all, or did everything come back empty?

Hint. *if the download stalls partway, delete data/<YOUR_GENE>.tsv before retrying. A partial cache file is worse than none, because the notebook will load it without complaint.*

Step 3 · Audit tool coverage (10 pts)

Group your variants by consequence class and count how many each tool could score. Report the table, the overall percentage scored by each tool, and explicitly which consequence classes have no missense score at all.

Read the result by its empty cells. A tool that returned nothing for a class of variant has told you nothing about them, which is not the same as telling you they are benign.

Step 4 · Apply a frequency filter you can defend (15 pts)

Choose a maximum credible allele frequency for your gene and justify it from the biology of this gene and the drugs it affects, not from habit.

A warning specific to pharmacogenes

Reduced-function pharmacogene alleles are frequently common. CYP2C19*2 sits around 15% in many populations and VKORC1 −1639G>A near 40% in some. A rare-disease filter of 0.1% would discard the most clinically important variants in these genes.

Whatever threshold you choose, you must inspect what it throws away. List the P/LP or drug response variants your filter removes. If it removes a known functional allele, either change the threshold or justify keeping it in your report. For contrast, also report what a 0.001 filter would have discarded.

Step 5 · Calibrate a predictor into evidence (15 pts)

Pick the predictor with the best coverage of your gene and calibrate it with calibrate(), as in Part 2 Section 8. Report the likelihood ratio and implied ACMG evidence strength per band.

Then look hard at the denominators. State how many labelled variants the calibration rests on, and flag any band resting on fewer than ten. A likelihood ratio computed on four variants is not evidence; it is noise with a decimal point.

Hint. *CADD is usually the right choice here because it scores every variant class, where REVEL is missense-only. Whichever you pick, say why.*

If your gene has too few labelled variants to calibrate at all, do not force it. Either fetch a second gene and pool, or state that calibration is not possible for your gene and explain why. Either is acceptable if you disclose it; neither is acceptable silently.

Step 6 · Classify the uncertain variants and export (5 pts)

Take the VUS rows, map each to the evidence strength of its calibrated band, and count how many move towards pathogenic, how many towards benign, and how many stay put.

The point of calibration

A variant whose score falls in a band with a likelihood ratio near 1 stays a VUS, because the predictor genuinely told you nothing. Only a large or small LR should move anything. Moving every high-scoring variant towards pathogenic is the misconception this whole exercise exists to correct.

Export to pgx_classified_<YOUR_GENE>.csv with at least: rsid, gene, consequence, aa_change, your score column, gnomad_af, evidence, provisional_call.

Step 7 · The AlphaGenome Atlas (code given; discussed in the report)

Run the Atlas cell on your gene. It prioritises non-coding variants, because that is where the Atlas earns its place, and caps the query so you stay inside your quota.

Read the attribution before the number. The top_feature column tells you why a variant scored highly, and it changes what the score means:

Top feature	What it means
MERGED_SPLICING	**A splicing mechanism. An RNA assay would settle it.**
ALPHAMISSENSE	**A coding mechanism. Your missense tools already saw this one.**
MAX_ABS_DNASE / CHIP_TF / RNA_SEQ	**A regulatory mechanism you can chase to a specific tissue.**
CACTUS_241_WAY / PHASTCONS_470_WAY	Conservation is carrying the score and the molecular mechanism is NOT resolved.

If you could not get a key, set SKIP_ATLAS = True and answer the Atlas part of the report from reasoning instead. No marks are lost for a disclosed skip.

Step 8 · Find your discordant variant (code given)

Run the discordance cell. It applies the Part 2 rules to your gene and flags variants where the tools disagree with each other, or with ClinVar, or where nothing could score them at all. Choose one variant from the output to write about in Step 9.

If nothing trips a rule, that is a legitimate result. Say so and explain why this gene's variants are concordant, rather than loosening the rules until something fires.

Step 9 · Report (3–4 pages, the 40%)

Answer these four in your report (PDF):

- **Thresholds and calibration.** Which frequency threshold did you choose and why is it right for this gene rather than a rare-disease gene? What did the 0.001 contrast discard? Which predictor did you calibrate and why? Report the likelihood ratios, state how many labelled variants they rest on, and say what that does to your confidence.
- **Coverage gaps and the regulatory genome.** Which consequence classes could no missense predictor score? Name one specific unscored variant and say what evidence would classify it and which tool or assay would produce that evidence. Then address the Atlas: what did it add, and what

did the attributions say was driving the top scores? If you skipped it, say what it could in principle have told you about which variants, and why nothing else in the course can.

- **A discordant variant, read mechanistically.** Take one variant from Step 8. State which rule it tripped, what each tool said, and what biology could produce that pattern. The question is not which tool is right; it is what would have to be true about this variant for both outputs to make sense. Argue for one explanation.
- Name two things your analysis cannot establish about your most interesting variant, drawing on the lecture's list: penetrance, mechanism and direction, zygosity and phase, tissue and timing, modifiers including phenoconversion, clinical utility. Then name the single experiment or dataset that would resolve your most interesting call, and what result would change your mind.

What to submit

- Kernel → Restart & Run All: confirm the notebook runs cleanly top to bottom.
- Commit the notebook, pgx_classified_<YOUR_GENE>.csv, your report (PDF), and a short README (your name, your gene, whether you ran the Atlas, and anything that didn't work) to your biot6900 repo. Don't commit the data/ folder: add it to .gitignore.
- Push, confirm the files appear on github.com, then post your repo link on Canvas.

Do not commit your AlphaGenome API key.

It belongs in your environment, never in the notebook. A key in a commit stays in the repository history even after you delete the line.

Due date & late policy

Due Thursday, October 22 at 6:00 pm (start of Week 7). Per the syllabus: –10% up to 1 day late, –20% up to 2 days, –50% up to 3 days, and no credit after the 3rd day unless we have arranged an alternative in advance.

How it's graded (100 pts · course 60 / 40 split)

Partial credit is awarded for a correct approach even with minor technical errors: if a step wouldn't run, say what you were trying to do and what happened. Explaining your reasoning earns credit.

What the marking rewards, and what it penalises

Earns credit: quoting your own numbers rather than the lab guide's; naming a specific variant rather than a category; disclosing a fallback and explaining its cost; committing to one interpretation and defending it.

Loses credit: treating a blank score as a benign result; quoting a likelihood ratio from a tiny band without caveat; calling a variant pathogenic on predictor scores alone; listing possible explanations instead of arguing for one.

Troubleshooting

Symptom	Likely cause & fix
My fetch returned only a handful of variants	**Check the gene symbol and the query string. Delete data/<YOUR_GENE>.tsv and run again. Do not continue with a partial file; every number after it will be wrong.**
The anchor allele for my gene is missing	**Your fetch is incomplete. Delete the cache file and retry. If it persists after two attempts, email me before continuing.**
CADD column is empty	**CADD is only served on the hg19 build by this API, so fetch_gene() makes a second request on that build and joins on the ClinVar variant ID. Check your connection and re-run the cell.**
revel is blank for most rows	**Expected, and it is the point. REVEL is missense-only. Confirm by checking the consequence column for those rows, and report it in Step 3 rather than treating it as an error.**
consequence looks wrong for a variant	**Check consequence_all. snpEff reports one effect per transcript and the code collapses them to the most severe, which can relabel a regulatory variant sitting inside an intron of another transcript.**
My gene has too few labelled variants to calibrate	**A real result, not a failure. Either pool with a second gene or state that calibration is not possible, and explain the consequence in your report.**

Symptom	Likely cause & fix
alphagenome ImportError	**pip install "alphagenome>=0.9.0", then Kernel → Restart. Needs Python 3.10 or newer. Or set SKIP_ATLAS = True.**
'No API key found in your environment'	**You set the variable but did not relaunch Jupyter from a new terminal. Close Jupyter, open a fresh terminal, start it from there.**
Atlas queries return errors	**Usually quota. Lower MAX_VARIANTS and try later, or set SKIP_ATLAS = True and disclose it.**
My numbers don't match the lab guide	**Expected. ClinVar changes constantly. The patterns should hold; the exact figures should not. Quote your own numbers in the report.**
Notebook won't run top to bottom	**You skipped a cell or ran them out of order. Later cells depend on variables defined earlier. Kernel → Restart & Run All, and fix the first error before moving on.**

Getting help

- Office hours: bring your laptop and your notebook.
- The Module 5 Lab Guide, Part 2: it walks through every technique used here, with a 'How to read it' block after each step.
- Email: include your name, the class title, and the exact error text (or a screenshot). Allow up to 48 hours for a reply, so don't wait until the night before.

BIOT 6900 · Module 5 Assignment (Lab 5) · Fall 2026

Computational Lab 5

Criteria	Ratings		Pts
Gene fetched and cached; three quality checks run and reported (execution)	**15 pts Full Marks**	**0 pts No Marks**	15 pts
Coverage audited by consequence class; unscoreable classes identified (execution)	**10 pts Full Marks**	**0 pts No Marks**	10 pts
Frequency filter applied; discarded variants inspected and the 0.001 contrast reported (execution)	**15 pts Full Marks**	**0 pts No Marks**	15 pts
Predictor calibrated correctly; sample size engaged with, or the small-gene route taken and disclosed (execution)	**15 pts Full Marks**	**0 pts No Marks**	15 pts
VUS classified against the calibrated bands and exported cleanly to CSV (execution)	**5 pts Full Marks**	**0 pts No Marks**	5 pts
Thresholds, coverage gaps and the discordant variant interpreted biologically, with evidence (intepretation)	**30 pts Full Marks**	**0 pts No Marks**	30 pts
Limits and next experiment discussed; fallbacks disclosed; report readable (communication)	**10 pts Full Marks**	**0 pts No Marks**	10 pts
			Total Points: 100